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Effects of ethnyl estradiol on the liver of male *Clarias batrachus*

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Abstract- The synthetic estrogen ethnyl estradiol (EE_2) is widely used in oral contraceptives. It is an endocrine disrupting contaminant of aquatic environments. The present study examines the effects of long- term exposure of EE_2 on liver tissue of male *Clarias batrachus*. Fishes weighing 100- 150 g (length :14 – 18 cm) were used and grouped into four different aquaria. Experimental groups were supplemented with increasing levels of EE_2 as 5µg, 10 µg and 20 µg per litre of EE_2 through Premarin (oral contraceptive pill). Control group fishes were not exposed to EE_2. After treatment for 90 consecutive days the histological alterations in the liver were studied by Transmission Electron Microscopy. Electron microscopic studies revealed disruptive changes in the hepatocytes as evident by enlarged nuclei, indistinct nucleoli, swollen and extended mitochondria and indistinct cell membranes after EE_2 exposure. Cell necrosis was clearly seen in EE_2 exposed fishes only. The present results show liver damage in male *Clarias batrachus* after EE_2 administration.

Key words: Fish, Liver, hepatocyte, Transmission Electron Microscopy

INTRODUCTION

Ethnyl estradiol (EE_2) is a kind of endocrine disrupting chemical (EDC) that alters the metabolic function of the body.[1] It is widely used in oral contraceptive pills. EE_2 enters into surface water through various sources like household, pharmaceutical, agricultural and animal wastes.

The presence of EE_2 in the surface water may have negative impacts on aquatic animals especially fish.[2-7] The exposure of fish to estrogen during early life, can induce permanent alteration of gonad development and physiological systems in fish.[8,9] Exposure to EE_2, especially targets the gonads. Also, the hepatic egg- yolk precursor protein vitellogenin, which acts as a biomarker, is induced in the liver of male fish as a response to estrogen or xenoestrogen exposure.[10-12]

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EE_2 exposure causes increase in epidermal growth factor receptor.[13] The epidermal growth factor receptor is linked with the actions of promoters, that can bring about formation of hepatic tumors by estrogens. Chronic exposure to EE_2, even through the use of oral contraceptive pills, has been shown to cause emergence of new hepatocyte populations. Oral contraceptives have high amount of EE_2. The leached EE_2 present in surface water is used for drinking purpose around the world. The amount of EE_2 exposure that produces adverse effects is still being researched.

The present research has been conducted on the catfish, *Clarias batrachus* which is popularly known as Magur in India. *Clarias batrachus* is hardy and is widely utilized in research.[14-18] Now a days, the rate of fish extinction is considerably higher than past. Decreasing fish populations affect the aquatic food chain and also the human communities that rely on fish as their main source

of protein.[19] EE_2 largely affects the fish populations. So, fishes are used as an indicator to this toxin in the water. Liver has been used as a target organ to evaluate the toxicity effects of various compounds due to its key role in xenobiotics transformation.

Therefore, the aim of the present study was to examine the effects of EE_2 exposure on the histological and ultra-structural changes in the liver of male catfish *Clarias batrachus*.

MATERIALS & METHODS

The experiment was conducted on the catfish *Clarias batrachus*. Healthy *Clarias batrachus* weighing 100-150 g (length: 14 – 18 cm) were purchased from "Hygienic Fish Market" Dhurwa, Ranchi in the monsoon season. The fish were transported in oxygenated bags and brought to the laboratory. They were kept in glass aquaria of 20" × 10" × 12" size, under normal aquatic conditions at a temperature of 23±5°C. They were acclimatized for a period of three days.

For preparation of stock solution, 100mg of EE_2 was dissolved in 100% ethanol. This solution was vortexed to dissolve EE_2 completely. Then the solution of EE_2 and ethanol was dissolved in 1000 ml (1 litre) of deionized distilled water. EE_2 was supplemented through Premarin.

The fishes were divided randomly into four different groups. Stocking density maintained for the experiment was ten fish per aquarium. They were maintained in four separate glass aquaria. Control group fishes were fed normal fish diet. Experimental groups were fed with diets containing 5 µg, 10 µg and 20 µg of EE_2 per litre supplemented through Premarin. Administration of EE_2 were done once a day for ninety continuous days.

After completion of ninety days the fishes from different groups were sacrificed. Their livers were quickly removed, cut immediately into small pieces of 1-2 mm in size and fixed in Karnovsky's solution (2.5% glutaraldehyde in 0.1 mol/l of phosphate buffer). The tissues were processed further for electron microscopic examination.

Transmission Electron Microscopy (TEM) studies were performed at Sophisticated Analytical Instrumentation Facility (SAIF) at the All India Institute of Medical Sciences (AIIMS), New Delhi.

RESULTS

Results on histological observations of hepatocytes are presented in Figures 1- 4. Hepatocytes of fishes exposed to EE_2 showed irregular arrangement with less distinct plasma membranes, nuclei and nucleolus as compared to those of control group which had very distinct plasma membranes and nuclei (Fig. 1-4). Exposure to high dose EE_2 resulted in the necrosis of the cell. The mitochondria were swollen and extended. Rough endoplasmic reticulum showed a concentric whorly appearance after exposure to medium dose EE_2, i.e., 10µg/L (Fig.- 3) whereas it appeared scattered in hepatocytes of fish exposed to high dose i.e, 20 µg/L (Figure- 4). In these hepatocytes the nucleus was pyknotic and the blood vessels appeared damaged leaving large empty spaces (Figure 4). Fat droplets and glycogen deposition was seen in the cells (Figure 2-4). Control group fishes showed normal hepatocyte structure (Fig. 1).

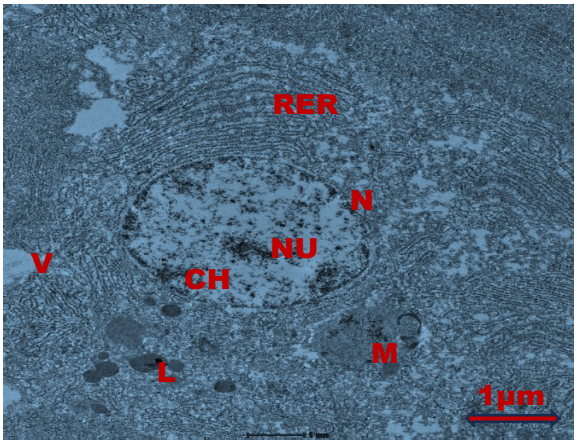


Fig.1- TEM of liver cells of fish of control group showing nucleus (N), nucleolus (NU), chromatin (CH), rough endoplasmic reticulum (RER), vacuole (V), mitochondria (M) & lysosome (L)

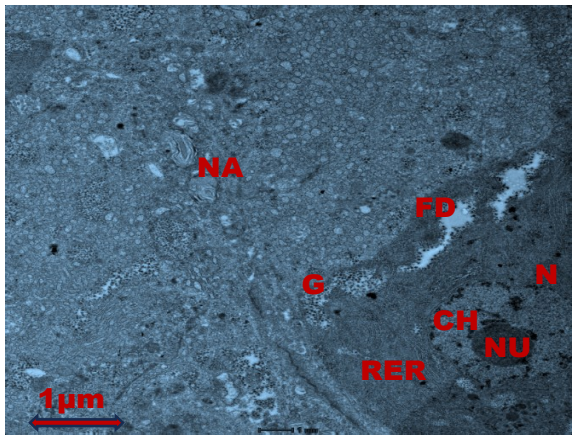


Fig.2- TEM of liver cells of fish exposed to 5µg/L EE_2 for 90 days showing nucleus (N), nucleolus (NU), chromatin (CH), rough endoplasmic reticulum (RER), glycogen (G), fat droplets (FD) & presence of necrotic area (NA)

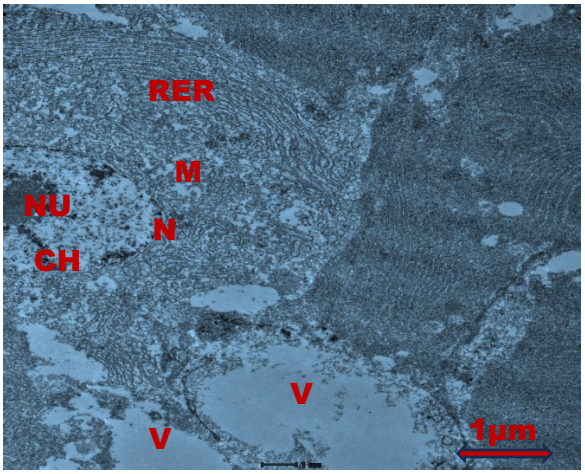


Fig. 3- TEM of liver cells of fish exposed to 10µg/L EE_2 for 90 days showing Nucleus (N), nucleolus (NU), rough Endoplasmic reticulum (RER), vacuole (V), chromatin (CH), and swollen mitochondria (M)

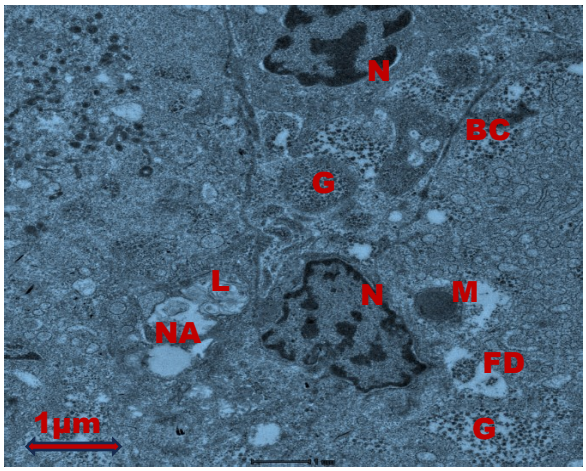


Fig. 4- TEM of liver cells of fish exposed to 20µg/L EE_2 for 90 days showing pyknotic nucleus (N), lysosome (L), swell-up mitochondria (M), deposition of glycogen (G), fat droplets (FD), bile canaliculi (BC), and presence of necrotic area (NA).

DISCUSSION

The present results clearly show that long- term exposure of *Clarias batrachus*, to EE_2 causes significant damage to its liver. The histological slides were observed under Transmission Electron Microscopy (TEM) which provided informative images that explained the changes in hepatocytes with and without EE_2 exposure. Necrosis was clearly evident in the liver cells of fishes exposed to the high dose of EE_2. The hepatocytes of these EE_2 exposed fishes showed less defined cell shape. Their arrangement was irregular and the plasma membranes were indistinct. The cells appeared to have different cytoplasmic density with swollen mitochondria. Blood vessels seemed damaged and empty spaces were visible. These changes were evident in the hepatocytes of fishes exposed to 20 µg EE_2/L. The present results enable us to suggest that administration of this dose of EE_2 for 90 days produces damage to the hepatic tissue of *Clarias batrachus*.

On the basis of the present results, it can be opined that increasing doses of oral contraceptive pills containing synthetic hormones may cause toxicity of liver cells. The activity of serum aspartate amino transferase (AST) and alanine amino transferase (ALT), may be increased leading to cellular degeneration of liver cells.[20] The results of present study show that synthetic estrogen affects liver function. It has been suggested that exposure to low doses of synthetic estrogen may cause destruction of red blood cells (RBC) in a dose dependent fashion.[21] This is so because liver functions in the filtration, storage and metabolism of blood. Liver also has a function in the formation and excretion of bile.[21] Our results are similar to previous reports, indicating that synthetic hormones cause hepatoxicity. These results are new for the catfish, *Clarias batrachus.* EE_2 is more potent than natural estrogens. Also, it remains in the blood for long duration after being administered thereby causing hepatoxicity.[22] Oral contraceptive pills contain progestogens together with estrogens. High dose of progestogens is metabolized to compounds, norethindrone acetate and norethisterone enanthate that affect liver function but to a lesser extent than the estrogens.[23]

Ultrastructural changes were also observed in the liver hepatocytes of *Clarias batrachus* exposed to even low dose of EE_2 i.e. 5µg/L. One other prominent effect observed was the absence of regular cytoplasmic compartmentalization. Similar effects have been observed on the hepatocytes of *Clarias gariepinus* which had been exposed to arsenic.[24] In the present study the mitochondria were observed to be swollen. Hyperplasia and hypertrophy of mitochondria have been observed after arsenic exposure to *Clarias gariepinus*.[24] In the present study hepatocytes of EE_2 exposed *Clarias batrachus* showed changes in the endoplasmic reticulum. The whorly appearance of endoplasmic reticulum was visible in the hepatocytes of fish exposed to medium -dose EE_2 (10 µg/L) but the endoplasmic reticulum became dispersed and scattered in high -dose (20 µg/L) exposed fishes. The present result is

similar to that of Sabullah *et al.* (2020)[25] but their study reported the effect of copper on the hepatocyte of *Puntius javanicus*. In the present study, EE_2 administration resulted in the enlargement of the nuclei accompanied by pyknosis. Enlargement of hepatocyte cell nuclei has been reported by Schwaiger *et al*., (2000)[26] in juvenile common carp (*Cyprinus carpio*) after EE_2 administration. However, they observed hypertrophy of the hepatocytes after EE_2 exposure whereas in the present study hepatocyte necrosis was evident after EE_2 treatment to *Clarias batrachus*.

The present TEM studies revealed increase deposition of glycogen and fat droplets in the hepatocytes in response to EE_2 treatment, as compared to the control fish. An opposite effect on glycogen deposition in hepatocytes has been observed by Haux and Norberg, (1985)[27] in estrogen-treated rainbow trout. They have reported a reduction in glycogen deposites.

The results of the present study suggest that EE_2 affects liver function in male *Clarias batrachus* and may induce liver damage.

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